ALPINA CHEMICALS S.P.A.

Via dell'Industria 14, 20090 Assago (MI), Italy
Tel +39 02 5550 1200 | quality@alpinachem.example

CERTIFICATE OF ANALYSIS

Certificate No. CoA-2026-005233 Issue date: 2026-10-02

Product and lot

Product	Citric Acid Anhydrous, Ph. Eur. / USP / FCC	**Lot / Batch No.**	L26-10-3402
Product code	CA-ANH-25	**CAS No.**	77-92-9
Manufacturing date	2026-09-14	**Retest / Expiry date**	2028-09-13
Lot quantity	24,000 kg	**Packaging**	25 kg multiwall paper bags, PE liner
EC food additive	E 330	**Country of origin**	Italy

Customer and shipment

Customer	Nordval Foods Plant 2 Zona Industriale Nord, 44122 Ferrara (FE), Italy	**Customer PO No.**	PO-77120-04
Delivery note	DN-2026-120377	**Shipped quantity**	12,000 kg (480 bags)
Dispatch date	2026-10-02	**Carrier**	Trasporti Padani S.r.l., vehicle FE 482 TX

Scope

This certificate reports the release analysis of the lot named above, performed by the quality control laboratory of Alpina Chemicals S.p.A. at the Assago site (ISO 17025 accredited, certificate LAB-0418). The results refer only to the sample tested and to the lot it represents. The lot is manufactured by fermentation of glucose syrup and purified by crystallisation; no animal-derived materials are used in any step.

Sampling

Samples were drawn on 2026-09-30 by the quality control sampler according to SOP QC-021 (ISO 2859-1, general inspection level II): 21 bags selected at random from the 960 bags of the lot, opened at the sampling station and taken with a stainless-steel thief to full depth. The primary samples were combined, homogenised in a riffle splitter and divided into three retained portions: one for the chemical and physical tests, one for the microbiological tests (sterile container) and one retained for two years after the expiry date.

Reading this certificate

Specifications are those of the Alpina Chemicals product specification AC-SPEC-CA-ANH rev. 11, which is aligned with the current European Pharmacopoeia monograph for anhydrous citric acid and with the USP and FCC monographs. A result shown as < is below the limit of quantification of the method. The analytical results are given on the next two pages, grouped by type of test: physical and identity, chemical purity, contaminants and microbiology.

Analytical results

A. Physical properties and identity

Test	Specification	Method	Result	Unit	Status
Appearance	White crystalline powder or colourless crystals	Visual	Conforms	-	Pass
Identification (A, B)	Complies	Ph. Eur. 2.3.1	Complies	-	Pass
Colour of solution	Not more intense than reference BY6	Ph. Eur. 2.2.2	Complies	-	Pass
Clarity of solution	Clear	Ph. Eur. 2.2.1	Clear	-	Pass
pH (50 g/L solution)	1.5 - 2.5	Ph. Eur. 2.2.3	1.9	-	Pass
Bulk density	0.75 - 1.00	Internal SOP QC-114	0.87	g/mL	Pass
Particle size, > 850 µm	max 5	Sieve analysis	1.4	%	Pass
Particle size, < 150 µm	max 10	Sieve analysis	2.9	%	Pass

B. Chemical purity

Test	Specification	Method	Result	Unit	Status
Assay (anhydrous basis)	99.5 - 100.5	Titration, Ph. Eur. 2.2.20	99.8	%	Pass
Water (Karl Fischer)	max 0.5	Ph. Eur. 2.5.12	0.19	%	Pass
Sulfated ash	max 0.05	Ph. Eur. 2.4.14	0.01	%	Pass
Oxalic acid	max 360	Ph. Eur. 2.2.25	62	ppm	Pass
Sulfate	max 150	Ph. Eur. 2.4.13	< 50	ppm	Pass
Chloride	max 50	Ph. Eur. 2.4.4	< 5	ppm	Pass
Readily carbonisable substances	Not more intense than reference	USP <221>	Complies	-	Pass
Calcium (Ca)	max 200	ICP-OES	< 10	ppm	Pass
Iron (Fe)	max 5	ICP-OES	< 1	ppm	Pass
Aluminium (Al)	max 0.2	ICP-MS	< 0.1	ppm	Pass
Residual methanol	max 3000	HS-GC, USP <467>	Not detected	ppm	Pass

C. Contaminants

Test	Specification	Method	Result	Unit	Status
Heavy metals (as Pb)	max 10	Ph. Eur. 2.4.8	< 5	ppm	Pass
Lead (Pb)	max 0.5	ICP-MS	< 0.1	ppm	Pass
Arsenic (As)	max 1	ICP-MS	< 0.1	ppm	Pass
Cadmium (Cd)	max 1	ICP-MS	< 0.05	ppm	Pass
Mercury (Hg)	max 1	ICP-MS	< 0.05	ppm	Pass

D. Microbiology

Test	Specification	Method	Result	Unit	Status
Bacterial endotoxins	max 0.5	Ph. Eur. 2.6.14	< 0.05	IU/mg	Pass
Total aerobic microbial count	max 1000	Ph. Eur. 2.6.12	< 10	CFU/g	Pass
Yeasts and moulds	max 100	Ph. Eur. 2.6.12	< 10	CFU/g	Pass
Enterobacteriaceae	Absent in 1 g	ISO 21528-2	Absent	-	Pass
Escherichia coli	Absent in 10 g	ISO 16649-2	Absent	-	Pass
Salmonella spp.	Absent in 25 g	ISO 6579-1	Absent	-	Pass

Statements

The lot complies with Commission Regulation (EU) No 231/2012 (specifications for food additives) and with the FCC monograph for citric acid. It is not genetically modified and does not need to be labelled under Regulations (EC) No 1829/2003 and 1830/2003. It has not been treated with ionising radiation and contains none of the allergens listed in Annex II of Regulation (EU) No 1169/2011. No BSE/TSE risk material is used.

Storage and handling

Store in the original closed packaging in a dry place at 5 - 30 °C, away from strong oxidising agents. Retest after 24 months from the manufacturing date. The product is hygroscopic: reseal opened bags.

Conclusion

The lot described in this certificate conforms to all the requirements of the specification AC-SPEC-CA-ANH rev. 11 and is released for sale.

Released by
Dr. Elena Ferretti
Head of Quality Control

Date
2026-10-02

Document
QF-114 rev. 7, 3 pages